



AGENTIC-AI @ UIUC · EDUCATION WORKSHOP · 2026

What students are building with agents

A student-run org on agentic AI — and the projects it produced.

01

How

did a student club grow to 500 members under 2 semesters?

THE PROBLEM

Courses teach concepts. Not practice.

Most learning experiences do not yet teach students how to collaborate with, evaluate, and direct AI agents.

THE MODEL

Lectures, then you ship.

Weekly lectures build the vocabulary. Then teams build a real, deployed agent.

01 Prompting, LLM, Memory Management

02 Tools, function calling, structured output

03 Retrieval-augmented generation (RAG) & Voice Agents

04 Multi-agent orchestration & evaluation

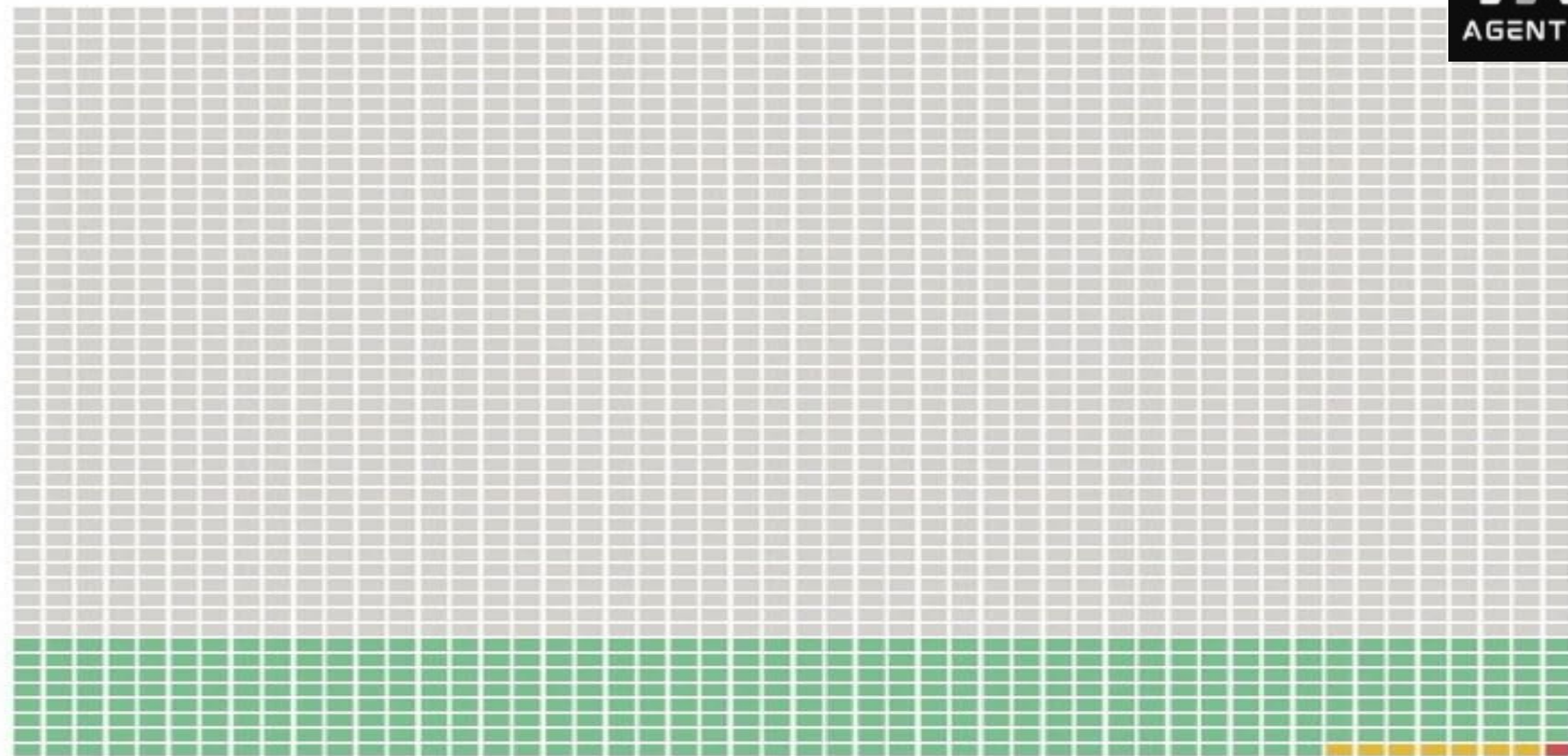
02

Why

teach this now?

Each dot is ~3.2 million people

2,500 dots = 8.1 billion humans. Color = most advanced AI interaction, Feb



Never used AI - ~6.8B (84%)

Free chatbot user - ~1.3B (16%)

Pays \$20/mo for AI - ~15-25M (~0.3%)

Uses coding scaffold - ~2-5M (~0.04%)

AN AGENT, IN THIS COURSE

**retrieves. reasons.
calls tools. acts.**

Not a chatbot that answers — a system that decides what to do and does it.

03

What

did the students build?

8

deployed agent projects built by student teams.

Equity, healthcare, voice, sports, food, music, portfolios, fabrication.

An agent that runs the lab.

Autonomous booking system for the campus fabrication shop laser cutters, CNC, 3D printers. A FIFO agent serves the next student, sends reminders, and an analytics chatbot answers staff questions about usage.

Python · discord.py · FastAPI · React · SQLite · OpenAI — replaces paid SaaS for ~\$10/mo



PROJECT 02 · MEDICAL VOICE AGENT

Call in. Get booked. No human needed.

A voice agent that identifies the patient, triages symptoms to a specialty by confidence score, finds open slots, and books, reschedules, or cancels — entirely over the phone.

Vapi.ai · FastAPI · Next.js · Supabase — tool-calling LLM over live phone audio

Two products. One corpus.

A source-grounded medical Q&A bot for patients — and, from the same knowledge base, a training simulator where med students interview an LLM patient and get scored on their diagnostic reasoning.

LangGraph · Qdrant · Groq Llama-3.3-70b · Flask — RAG state machine + agent layer

Find clubs and food with AI

A searchable AI-powered catalog for UIUC clubs that automatically gathers organization info, events, and free food opportunities from Instagram, Discord, Telegram, WhatsApp, and websites.

LangChain · ChromaDB · GPT-4o · Streamlit — multi-agent retrieval + valuation

AND FOUR MORE

**Sports Analyst
Agent**

Equity Agent

**Music
Recommendation
System**

**Portfolio
Manager**

THE COMMON SHAPE

RETRIEVE → **DECIDE** → **ACT**

grounded in real data

an LLM picks the tool

a real side effect

Every team built the same loop against a different domain.

The lecture is the setup. The repo is the exam.

Four lectures of vocabulary, then a deployed agent. The portfolio is public on GitHub.

